

SOLANDER ISLAND, BC, CanadaWMO: **714790**Lat: **50.111N**Lon: **127.940W**Elev: **94**StdP: **100.21**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.4	0.9	-9.2	1.7	1.5	-6.0	2.3	3.8	33.0	5.7	29.8	6.2	8.5	10	1.018

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.6	17.4	14.2	16.1	14.0	15.4	13.7	15.4	16.6	14.8	15.7	14.3	15.2	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	10.7	15.6	14.4	10.4	15.1	13.9	10.0	14.6	43.3	16.8	41.6	15.8	40.4	15.2	19.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.8	25.3	21.9	DB	-2.2	22.7	3.0	3.3	-4.4	25.1	-6.1	27.0	-7.8	28.9	-10.0	31.3	(4)
(5)			WB	-4.4	16.7	2.9	1.2	-6.5	17.5	-8.2	18.2	-9.8	18.9	-11.9	19.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	6.2	5.6	6.0	7.3	9.2	10.9	12.3	13.2	12.9	10.3	7.9	6.0	(6)	
	DBStd	3.43	2.31	2.21	2.22	1.87	1.94	2.01	2.01	1.97	1.71	1.77	2.32	2.24	(7)	
	HDD10.0	721	118	122	127	83	38	12	7	4	1	17	69	123	(8)	
	HDD18.3	3406	376	356	383	329	284	224	186	159	162	250	314	381	(9)	
	CDD10.0	358	1	1	2	3	12	38	79	103	89	25	5	1	(10)	
	CDD18.3	2	0	0	0	0	0	1	0	1	1	0	0	0	(11)	
	CDH23.3	7	0	0	0	0	0	2	3	2	1	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	9.9	11.8	10.4	10.3	9.4	9.0	8.4	9.1	8.8	8.6	10.0	11.0	11.6	(14)	
(15) Precipitation	PrecAvg	2724	343	234	263	186	114	90	75	107	159	355	407	390	(15)	
	PrecMax	3546	599	574	456	413	188	152	210	242	317	641	596	692	(16)	
	PrecMin	2176	163	78	75	54	36	33	17	23	37	101	224	168	(17)	
	PrecStd	320	99	103	81	68	37	31	45	55	71	131	122	105	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	12.1	14.0	16.0	17.2	19.6	18.2	20.3	19.7	15.7	13.7	11.4	(19)	
		MCWB	8.4	7.7	7.7	9.6	11.7	12.9	14.9	15.0	14.9	12.0	10.8	8.8	(20)	
	2%	DB	10.2	10.0	10.6	12.2	13.7	15.0	16.1	16.9	16.8	14.2	12.3	10.4	(21)	
		MCWB	8.5	7.8	8.1	9.3	10.8	12.8	14.1	14.8	14.2	12.3	11.0	8.7	(22)	
	5%	DB	9.5	9.1	9.5	10.4	12.2	13.7	15.2	16.0	15.6	13.2	11.5	9.5	(23)	
		MCWB	8.2	7.5	8.0	8.5	10.2	12.2	13.8	14.5	13.9	11.9	10.3	8.4	(24)	
	10%	DB	8.9	8.4	8.7	9.5	11.3	12.9	14.5	15.2	14.9	12.6	10.7	8.9	(25)	
		MCWB	7.8	7.1	7.4	8.0	9.9	11.7	13.4	14.1	13.4	11.5	9.5	7.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	10.0	10.1	11.3	12.5	14.4	15.7	16.2	16.2	14.4	12.8	10.6	(27)	
		MCDB	10.7	10.5	10.3	13.2	15.5	16.5	18.1	17.3	18.2	14.7	13.0	10.8	(28)	
	2%	WB	9.3	8.9	9.2	10.1	11.4	13.4	14.8	15.5	15.1	13.2	11.5	9.6	(29)	
		MCDB	9.6	9.4	9.9	11.7	13.0	14.8	15.7	16.5	16.3	13.5	11.7	10.0	(30)	
	5%	WB	8.7	8.2	8.5	9.2	10.7	12.7	14.4	15.0	14.3	12.5	10.6	8.8	(31)	
		MCDB	9.1	8.7	9.4	10.1	12.0	13.7	15.2	15.9	15.2	12.8	10.9	9.3	(32)	
	10%	WB	8.3	7.5	7.7	8.4	10.2	12.2	13.9	14.5	13.8	11.8	9.8	8.1	(33)	
		MCDB	8.7	8.1	8.7	9.3	11.3	13.1	14.6	15.3	14.5	12.3	10.3	8.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.7	2.9	3.2	3.0	2.6	2.5	2.4	2.6	2.7	2.7	2.8	2.8	(35)	
		MCDBR	3.3	3.6	3.9	4.3	4.2	3.8	3.7	4.0	4.0	3.5	3.2	3.3	(36)	
	5% WB	MCWBR	2.4	2.5	2.9	2.9	2.8	2.6	2.5	2.7	3.0	2.9	3.2	3.1	(37)	
		MCWBR	2.7	2.9	3.3	3.9	3.7	3.4	3.6	3.8	3.6	3.0	3.0	3.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.321	0.325	0.345	0.375	0.380	0.366	0.359	0.362	0.353	0.342	0.327	0.316	(40)	
		taud	2.444	2.505	2.460	2.360	2.385	2.461	2.526	2.526	2.534	2.508	2.431	2.397	(41)	
	Ebn at Noon		720	823	863	869	878	892	893	875	846	787	701	662	(42)	
		Edh at Noon	62	74	91	112	114	106	98	94	84	73	62	56	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.78	1.66	2.62	4.05	5.08	5.41	5.46	4.79	3.46	1.90	0.94	0.64	(44)	
	RadStd		0.10	0.25	0.32	0.30	0.47	0.53	0.56	0.42	0.28	0.25	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-290	+106	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page